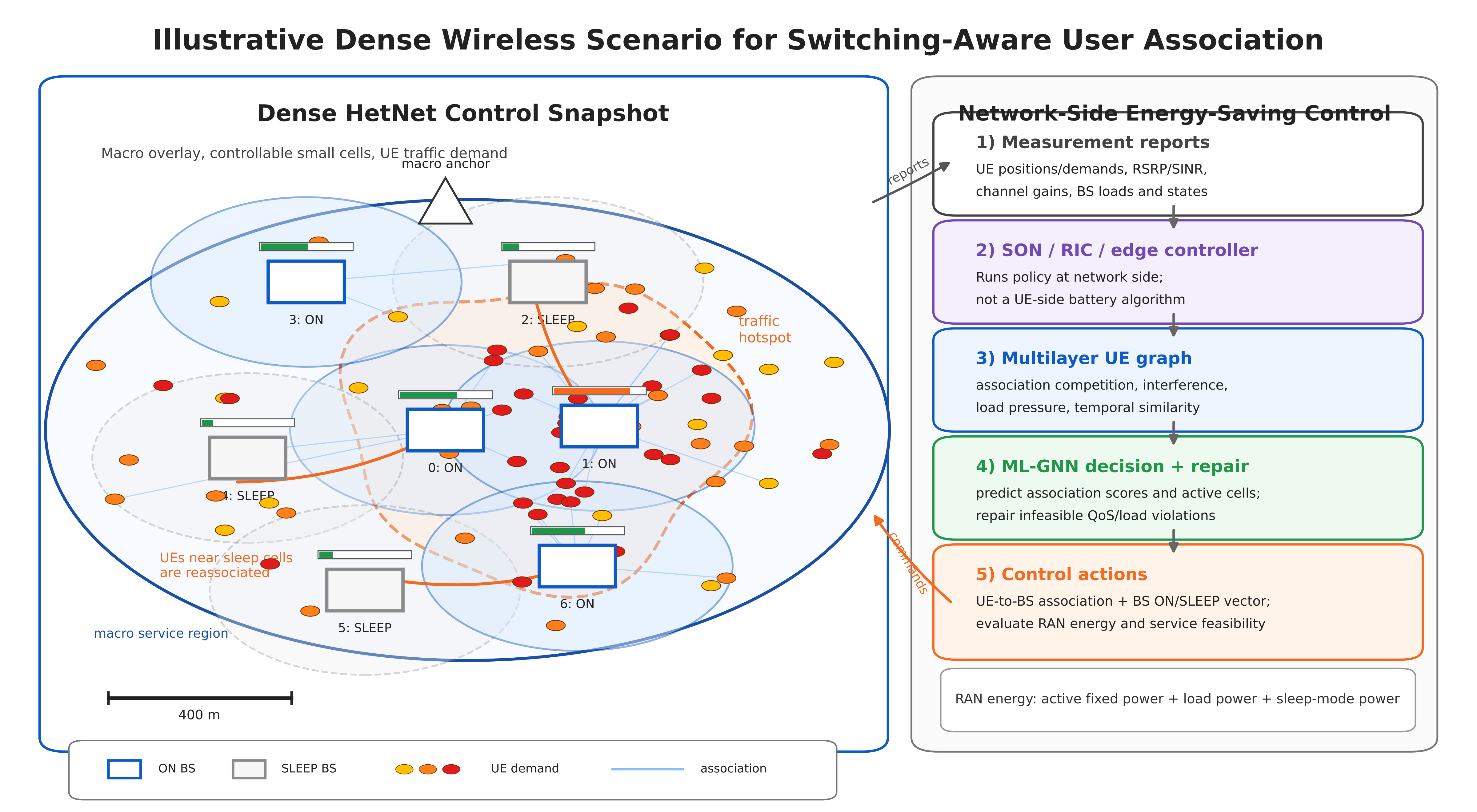

Illustrative Dense Wireless Scenario for Switching-Aware User Association
Dense HetNet Control Snapshot
Macro overlay, controllable small cells, UE traffic demand
macro anchor
3: ON
2: SLEEP
traffic hotspot
0: ON
1: ON
4: SLEEP
UEs near sleep cells are reassociated
5: SLEEP
6: ON
macro service region
400 m
ON BS
SLEEP BS
UE demand
association
reports
commands
Network-Side Energy-Saving Control
1) Measurement reports
UE positions/demands, RSRP/SINR, channel gains, BS loads and states
2) SON / RIC / edge controller
Runs policy at network side; not a UE-side battery algorithm
3) Multilayer UE graph
association competition, interference, load pressure, temporal similarity
4) ML-GNN decision + repair
predict association scores and active cells; repair infeasible QoS/load violations
5) Control actions
UE-to-BS association + BS ON/SLEEP vector; evaluate RAN energy and service feasibility
RAN energy: active fixed power + load power + sleep-mode power